Econ490: Financial Data Science

Data Science for Macro and Financial Analysis
Department of Economics, Duke University

Instructor Information

Prof. Aguilar ‖ mike.aguilar@duke.edu ‖ https://www.linkedin.com/in/mike-aguilar-econ/
Office Hours: see Canvas; or by appointment

Class Schedule and Delivery

Synchronous, In-Person ‖ Social Science 107
24Aug2026-4Dec2026 ‖ T/R 11:45AM-1:00PM

Pre-Requisites

ECON104D - Econometrics and Data Science
Some familiarity with R and/or Python
Some background in financial economic concepts

Course Description

Financial Data Science is the process of transforming financial data into useful products through statistical modeling, visualization, and software engineering.

This course provides a comprehensive, hands-on introduction to the full data science pipeline as applied to financial and macroeconomic analysis. Students will learn to 1) scope analytic problems, 2) acquire and clean data, 3) engineer features, 4) conduct exploratory data analysis (EDA), 5) construct and evaluate models, 6) visualize results and 7) deploy insights.

Learning Objectives

- Financial Data: Locate, clean, merge, and interpret the major formats of financial data, including market data (OHLCV, returns, bid/ask spreads), fundamental data (financial statements, earnings estimates), macroeconomic data (inflation, yields, employment), and alternative data (news, sentiment, etc..)
- Computing: Use modern data science tools to build reproducible financial applications leveraging tools like R and Python, Git, and APIs
- Visualization and Communication: Design and create appropriate static and interactive visualizations, dashboards, and data storytelling
- Exploratory Data Analysis (EDA): Design and implement appropriate methods to support choice of necessary statistical learning techniques, including univariate statistical distributions, correlations, time dependence, and feature engineering
- Statistical Learning: Design and implement appropriate methods to achieve project goals, including dimension reduction, clustering, regression, classification, and forecasting
- Uncertainty: Implement and interpret techniques such as bootstrapping and Monte Carlo analysis
- Financial Reasoning: Able to explain and apply financial economic concepts such as risk and return, diversification, factor exposures, yields, and volatility

Communication
Assignments, announcements, grades, readings, and other information will be posted on Canvas. The Canvas site will be the primary method of communication for this course, so please check it frequently.

Course Materials
There is no required textbook for this course. Optional readings will be made available throughout the term. Some initial suggestions include:

- Financial Data Science with Python, by Haojun Chen
- Financial Data Science, by Calafiore, El Ghaoui, Fracastoro, and Tsai

Technology

- Cell phones and similar devices: Please do not use cell phones, or similar devices, during class unless otherwise instructed to do so.
- Laptops and tablets: Please bring your laptop to every class. You are welcome to take notes on a tablet. However, the work will be doing together is most conducive to a laptop. Specifically, we will be using R regularly. Be sure to have R available on your laptop. Class demonstrations and solutions will be completed on a Windows OS, but MAC variations are possible.
- Canvas: Course materials such as data, slides, and assignments, will be distributed through the Canvas course site.

Honor Code
I take academic integrity seriously. All students are expected to adhere to the Duke Community Standard. It is the responsibility of each student to understand and follow Duke policies regarding academic integrity. Examples of academic dishonesty include, but are not limited to, cheating on an exam, paraphrasing or summarizing any source without adequate documentation, handing in someone else's work and claiming it is yours (including services such as Coursehero), collaborating with another student inside or outside this class on graded work that is to be done independently, recycling your work or work of other students from other classes. Generative AI may not be used for any class activities or assignments unless otherwise instructed. If I see evidence of any academic dishonesty, I will confront it according to the procedures described in the Bulletin of Duke University.

The Duke Community Standard says:
Duke University is a community dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, and accountability. Citizens of this community commit to reflect upon these principles in all academic and non-academic endeavors, and to protect and promote a culture of integrity. To uphold the Duke Community Standard, students affirm their commitment to uphold the values of the Duke University community by signing a pledge that states:

- I will not lie, cheat, or steal in my academic endeavors
- I will conduct myself honorably in all my endeavors
- I will act if the Standard is compromised

Grading

I will use the categories outlined below to develop a numeric score. Numeric scores will be translated into letter grades using final course-grade thresholds determined by the instructor, taking into account the overall difficulty and performance of the class.

Course Project (90%): The vast majority of your course grade will center around a comprehensive capstone-style project. Pretend you are an entrepreneurial financial data scientist. Your task is to develop a product. Specifically, you should identify a macro-financial problem, and then design, implement, and deploy a solution, which you will communicate and critically evaluate.

To ensure you remain on track, your grade will be reflected through weekly deliverables. Each deliverable is designed to support your product development and is timed to correspond to the desired project life cycle.

Class Participation (10%): Participation is judged in several ways.

- Attendance: Active participation is essential in this course. If you miss class, you are responsible for staying up to date and reviewing notes from classmates. Absences due to illness, emergency, or university-related obligations are excused when you submit an Incapacitation Form or Dean's Excuse. Excessive absences, excused or unexcused, may impact your participation grade.
- In-Class Exercises: Most class sessions will culminate with a hands-on exercise that will apply the skills we are exploring that day. I will guide the class through parts of the exercise, and with time permitting, will allow you to work in small groups. These submissions are graded for effort and engagement, not for correctness, and you are welcome to submit partial code/documents as long as it reflects a genuine attempt to work through the exercise.
- Engagement: Being engaged with the course material is vital for success. A few examples include: Coming to class prepared by completing any necessary readings and pre-work, asking questions in and out of class after having thought carefully about the issue, answering questions I pose to the class, asking your classmates thoughtful questions during presentations, etc..
- Norms: I expect professional behavior and adherence to the Duke Community Standard at all times. Cells phones and similar devices should not be used during class, unless explicitly told to do so. Other examples include, but are not limited to: Being on time and being respectful of classmates. Violations are a forfeiture of your entire Class Participation grade and may be reported to the Office of Student Conduct and Community Standards.

Tentative Schedule

The term *R&D* stands for Research and Development. We will spend the first half of the semester doing Research and the second half in Development.

First Half:

- Week 1: Course expectations and key finance concepts review
- Week 2: Financial data types and sources
- Week 3: Data acquisition via APIs, Git, and code architecture best practices
- Week 4: EDA and Unsupervised Learning
- Week 5: Unsupervised and Supervised Learning
- Week 6: Supervised Learning and Forecasting
- Week 7: Data interfaces such as dashboards and LLMs

Second Half:

The rest of the semester generally will be a mix of status updates, labs, and mini lectures.

- Status Updates: each student (schedule permitting) will get time to present product status updates and gain feedback from their classmates
- Mini Lectures: discussion of data science, finance concepts, or current events that are relevant to products being developed by the students
- Lab: Students will work on product development under the guidance of the instructor

Other Disclosures:

Modifications: The instructor reserves the right to alter any of the aforementioned components of this syllabus or other class-related materials in an effort to enhance the pedagogical experience.

Salutation: While we are working together in this official capacity, my preferred salutation is Prof. Aguilar. Once our work together is complete, I welcome less formality.

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Generative AI may not be used for class activities or assignments unless its use is explicitly permitted by the instructor for that activity or assignment.

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Academic Accommodations: If you are a student with a disability and need accommodations for this class, it is your responsibility to register with the Student Disability Access Office (SDAO) and provide them with documentation of your disability. The SDAO will work with you to determine what accommodations are appropriate for your situation. Please note that accommodations cannot be provided until the instructor receives a Faculty Accommodation Letter from the SDAO. Contact: sdao@duke.edu

Resources to Support Student Success: If you are having trouble completing assignments or understanding the materials, please consult with me about appropriate course preparation and readiness strategies as needed. Either send me an email or visit office hours to discuss the personal or academic difficulties you are facing. I may also direct you to other resources on campus.

The Academic Resource Center (ARC) offers services to support students academically during their undergraduate careers at Duke. The ARC can provide support with time management, academic skills and strategies, course-specific tutoring, ADHD/LD coaching, and more. ARC services are available free to any Duke undergraduate students studying any discipline. Contact: (919) 684-5917 or theARC@duke.edu

Mental Health and Wellness Resources: Student mental health and wellness are of primary importance at Duke, and the university offers resources to support students in managing daily stress and self-care. Some resources are listed below:

DuWell provides Moments of Mindfulness (stress management and resilience building) and meditation programming (Koru workshop) to assist students in developing a daily emotional well-being practice. All are welcome and no experience is necessary.

If your mental health concerns and/or stressful events negatively affect your daily emotional state, academic performance, or ability to participate in your daily activities, many resources are available to help you through difficult times.

DukeReach provides services for students who are experiencing significant challenges relating to mental health, physical health, social adjustment, and/or a variety of other stressors.

Counseling and Psychological Services (CAPS) services include individual and group counseling services, psychiatric services, and workshops. To initiate services, walk-in/call-in 9:00 AM–4:00 PM (M/W/Th/F) and 9:00 AM–6:00 PM Tuesdays. CAPS also provides referral to off-campus resources for specialized care. Contact: (919) 660-1000.

TimelyCare (formerly known as Blue Devils Care) is an online platform that is a convenient, confidential, and free way for Duke students to receive 24/7 mental health support through TalkNow and scheduled counseling.

Financial Support: If you are having difficulty with textbook, software, or supply costs associated with this course, here are some resources:

- Contact the financial aid office, whether you are on aid or not. They have loans and resources for connecting students with programs on campus that might be able to help alleviate these costs.
- DukeLIFE offers course materials assistance for eligible students. Please note that students who are eligible for DukeLIFE benefits are notified prior to the start of the semester; program resources are limited. Students who have limited access to computers may request loaner laptops through the DukeLIFE Technology Assistance Program. Please note that supplies are limited.
- Duke Libraries offers textbook rentals through the Top Textbooks program, where you can rent a textbook for 3 hours at a time.

Discussion Guidelines: Civility is an essential ingredient for academic discourse. Differences in beliefs, opinions, and approaches are to be expected. Active interaction with peers and your instructor is essential to success in this course, so please adhere to these guidelines:

- Respect that others' opinions and beliefs may differ from your own. If you disagree, you may critique the idea, but not the person.
- Listen carefully, be courteous, and do not interrupt.
- Support your statements with evidence and a rationale.
- Try to moderate how you contribute to the discussion—if you have a lot to say, try to avoid dominating the conversation; if you are reluctant to speak up, try to find an opportunity to share your perspective.

Please bring any communications you believe to be in violation of this class policy to the attention of your instructor.